

Eshupriye Belgotra

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Education

Bachelor of Science in Computer Science | University of Victoria

Jan 2023 – Apr 2028 Victoria, BC

- GPA: 7.38/9.0 ($\approx$ 3.70/4.0)
- Coursework: Algorithms & Data Structures, Object-Oriented Programming, Software Development Methods, Systems Analysis, Human-Computer Interaction, Functional Programming

Technical Skills

Languages: Python, Java, C, JavaScript, TypeScript, SQL, Swift, Bash, OCaml, Ruby, Racket

Frameworks & Tools: Flask, FastAPI, React, Node.js, REST APIs, WebSockets, Git, Linux/Unix

AI/ML: RAG architecture, vector embeddings, semantic search, NLP, scikit-learn, HistGradientBoosting, reinforcement learning, prompt engineering

Databases & Cloud: PostgreSQL, MySQL, SQL Server, Azure DevOps, SSIS, SharePoint, Redis

Experience

RBC Borealis Let's Solve It | Spring 2026 Cohort

Mar 2026 – May 2026 Victoria, BC

- Conducted exploratory data analysis on **1,500+ longitudinal youth health engagement records** spanning 3+ years in a team of 4 mentees alongside 2 senior AI researchers, surfacing disengagement signals across **20+ behavioural features** using Python, Pandas, and NumPy
- Reduced high-dimensional engagement data to 2D embeddings by implementing UMAP clustering in Python and scikit-learn, isolating **4+ distinct user cohorts** to inform downstream model design
- Engineered an ML pipeline to flag at-risk youth navigating chronic illness by training PyTorch models on 3+ years of longitudinal engagement data, enabling earlier clinical intervention
- Designed an analytics dashboard serving **4+ clinical and program teams** by visualizing retention curves, engagement trends, and disengagement risk flags using Python, Matplotlib, and GA4

Clinical Application Analyst (Co-op) | Island Health

Sep 2024 – Apr 2025 Victoria, BC

- Designed and built a multi-screen Clinical Governance Membership Management System in Power Apps, replacing manual spreadsheets and centralizing membership tracking for **5,000+ members** across BC's provincial healthcare system, with automated permission workflows, real-time roster edits, and audit-ready compliance reporting
- Engineered automated data pipelines querying and transforming **10,000+ patient records** in SQL Server, reducing manual data-entry time by **80%** and improving cross-system data accuracy
- Migrated legacy mental health housing system from MS Access to SQL Server; designed schema, authored data-loading scripts, and corrected fiscal calculation logic to support **2,000+ annual residential applications**
- Built PowerShell automation scripts and SSIS pipelines for recurring data migration and system maintenance, saving the team **10+ hours per week**
- Delivered **3 production releases** via Azure DevOps and ServiceNow with **zero critical incidents** through end-to-end QA; resolved bugs in MAMDA Virtual Care Account System affecting user-syncing and account-visibility logic

Projects

Personal Portfolio Site [\[eshu.earth\]](#)

Next.js · Tailwind · Supabase · Redis · Three.js · WebAudio · OpenAI · GitHub Actions

May 2026

- Built a RAG-backed AI chat assistant using OpenAI embeddings and Supabase pgvector, retrieving context from 38 embedded knowledge chunks with sub-second latency
- Automated knowledge base sync with a GitHub Actions pipeline — parsing the LaTeX resume with Claude, re-embedding with OpenAI, and upserting to Supabase on demand
- Engineered a low-latency TTS pipeline by streaming and chunking AI responses in real time, reducing time-to-first-audio to under 500ms
- Integrated a real-time lip-synced **3D** avatar using Three.js and the WebAudio API, animating mouth shapes from live audio amplitude at **60fps**
- Hardened the API with per-IP rate limiting, audit logging, and a GDPR-compliant privacy policy covering **5** jurisdictions

PreCheck.ai [\[precheckai.tech\]](https://precheckai.tech)

nwHacks 2026 Finalist: Best Use of Gemini API & Best Use of .tech 600+ participants

Jan 2026

- Architected a **4-stage ML validation pipeline**: Gemini Vision OCR, RAG-based policy retrieval against MSP/Pacific Blue Cross/Sun Life, HistGradientBoosting risk scoring (0–100 denial probability), and a remediation dashboard with citation-level fix instructions
- Built full-stack MVP in **24 hours** using React, TypeScript, Vite, FastAPI, Gemini API, and scikit-learn; human-in-the-loop design ensures AI suggests and clinic admin approves before any submission

SkyNet Space Biology Knowledge Engine [\[github.com/eshu3104/nasa-hackathon-2025\]](https://github.com/eshu3104/nasa-hackathon-2025)

NASA SpaceApps Hackathon 2025

Oct 2025

- Built an AI semantic search platform over **608 NASA publications (11,460 chunks)** using OpenAI embeddings, cosine similarity search, and Python/Flask, delivering role-differentiated research insights for scientists, educators, and funding managers
- Integrated GPT-4o-mini document summarization pipeline, reducing literature review time by **60%**; designed context-aware follow-up query interface with prompt engineering for deeper exploration

AWS DeepRacer [\[github.com/eshu3104/AWSDeepRacer\]](https://github.com/eshu3104/AWSDeepRacer)

Reinforcement Learning Competition

Nov 2024

- Engineered a reward function for an RL racing model optimizing speed, steering, and track positioning through **200+ training iterations**; placed **5th out of 100 competitors** with a race time of **9.729s**, within 2 seconds of the world record on the re:Invent 2018 track

Awards, Certifications & Languages

Awards: nwHacks 2026 Finalist (Best Use of Gemini API & .tech), FoodBridge People's Choice Award (UVicHacks 2026), Health & Safety Checker 5th Place (Tenfold Cursor Hackathon, 2025), Jarmila Vlasta Von Drak Thouvenelle Co-op Scholarship (UVic, Feb 2025)

Certifications: CS50x: Introduction to Computer Science (HarvardX, 2021), Python for Data Science & ML (Udemy, 2021), Complete Python Bootcamp (Udemy, 2020)

Languages: English (Native), Hindi (Native), French (Working Proficiency)